Scope of exam:

Q1 intro to electrical engineering

Q2 resistive circuits + Circuit Analysis Theorems and Techniques

Q3 resistive circuits + Circuit Analysis Theorems and Techniques

Q4 Capacitor and Inductor part 1

Q5 Capacitor and Inductor part 2 + Steady State Sinusoidal Analysis part 1

Formula given in exam:

Current through capacitor, $i_C(t) = \mathrm{C}\frac{dv_C(t)}{dt}$ Voltage across inductor, $v_L(t) = \mathrm{L}\frac{di_L(t)}{dt}$
Complete solution for the transient response of first-order circuit: $x(t) = [x(0) - x(\infty)]e^{-t/\tau} + x(\infty)$ $\tau_{RC} = RC$; $\tau_{RL} = L/R$
Standard form of second-order characteristic equation: $s^2 + 2\alpha s + \omega_0^2 = 0$ $\alpha_{parallel} = \frac{1}{2RC}$; $\alpha_{series} = \frac{R}{2L}$; $\omega_0 = \frac{1}{\sqrt{LC}}$
Complete solution for the transient response of second-order circuit: Overdamped: $x(t) = x(\infty) + A_1e^{s1t} + A_2e^{s2t}$ $\alpha > \omega_0$; s1 ≠ s2 Critically damped: $x(t) = x(\infty) + (A_1 + A_2t)e^{-\alpha t}$ $\alpha = \omega_0$; s1 = s2 Underdamped: $x(t) = x(\infty) + (A_1\cos\omega_d t + A_2\sin\omega_d t)e^{-\alpha t}$ $\alpha < \omega_0$; s = $-\alpha \pm j\omega_d$ $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$
For a sinusoidal voltage in time domain: $x_1(t) = X_1\cos(\omega t + \theta_1)$ Its phasor (phasor domain) is defined as: $X_1 = X_1 \angle \theta_1$

Practice questions

1. Three light bulbs are connected in series to a 120-V source as shown in Fig. 1. Find the current *I* through the bulbs. Each bulb is rated at 120 volts. How much power is each bulb absorbing? Do they generate much light? How about when the bulbs are connected in parallel to the 120-V source, calculate the current through each bulb.

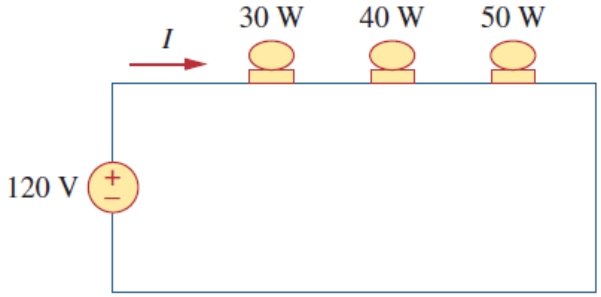


Fig. 1

2. Calculate the node voltages v1, v2, and v3 in the circuit of Fig. 2.

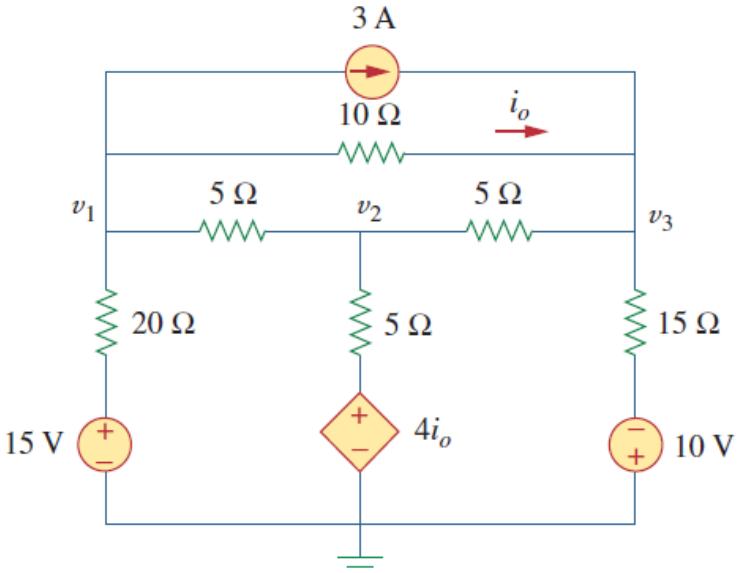


Fig. 2

3. Calculate the mesh currents *i*1 and *i*2 in Fig. 3.

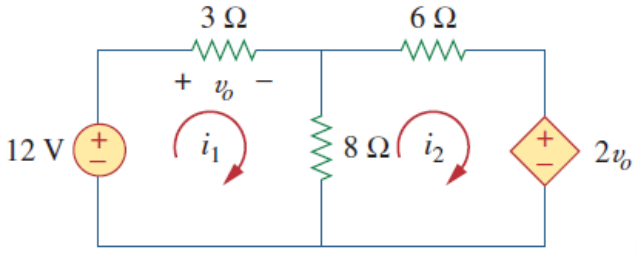


Fig. 3

4. Use superposition to find *Vo* in the circuit of Fig. 4.

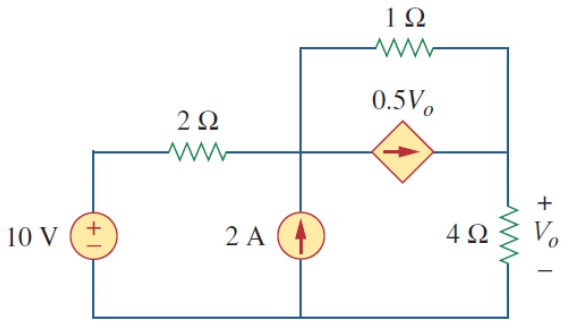


Fig. 4

5. Determine the Norton equivalent at terminals a-b for the circuit in Fig. 5.

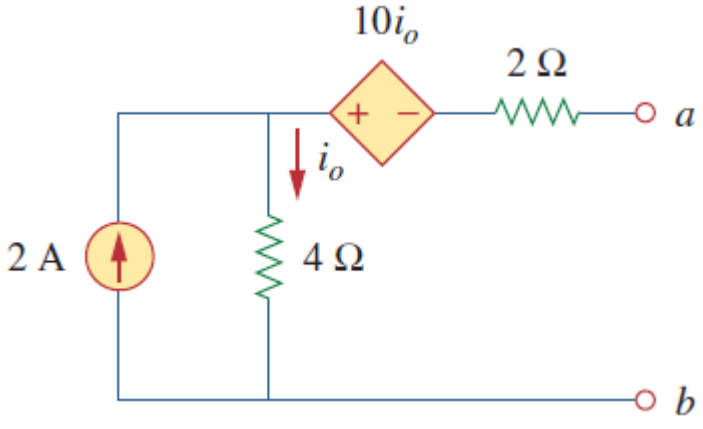


Fig. 5

6. The current through a 5-mH inductor is shown in Fig. 6. Determine the voltage across the inductor at t = 1, 3, and 5 ms.

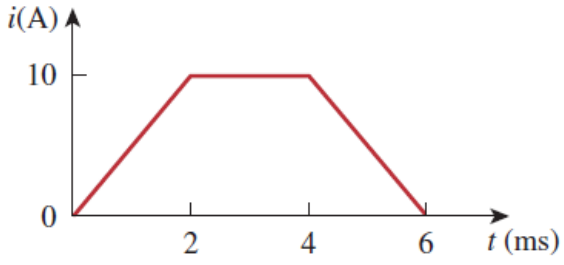


Fig. 6

7. Obtain *v*(*t*) and *i*(*t*) in the circuit of Fig. 7.

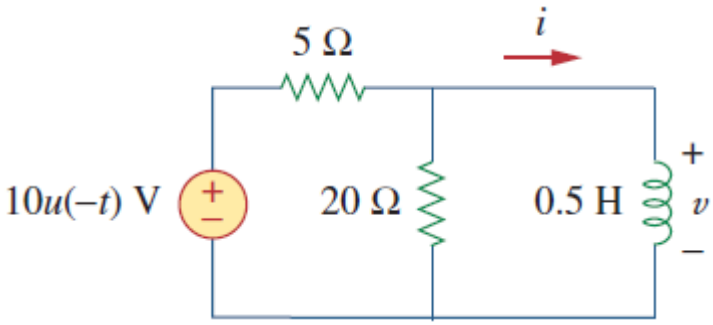


Fig. 7

8. Determine $i(t)$ for t 7 0 in the circuit of Fig. 8.

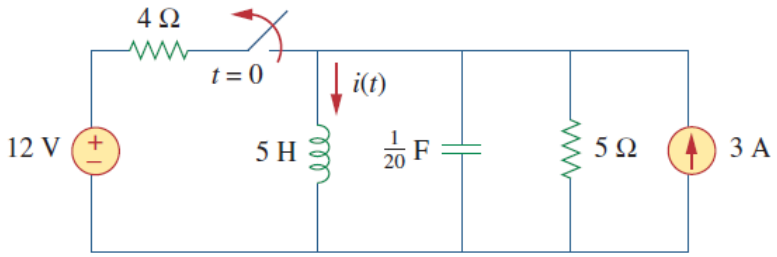


Fig. 8

9. Calculate $i(t)$ in the circuit of Fig. 9.

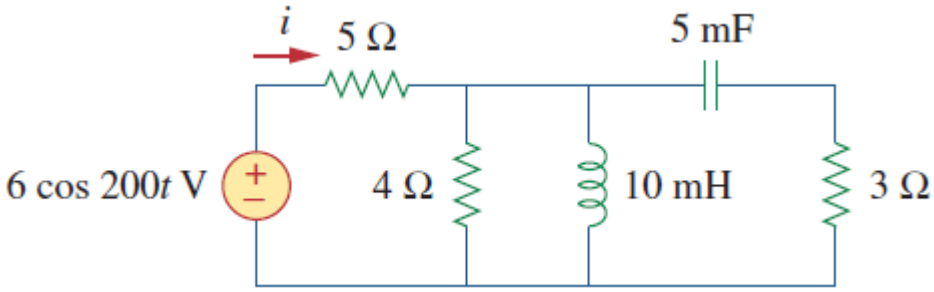


Fig. 9